



MINERAL TITLE, RE-IMAGINED

LANDroid

A visual ownership engine for the working landman.

Landmen still ship title work in Excel.

- **Spreadsheet sprawl**

Owner tables, snapshot grids, and royalty math live in fragile workbooks.

- **Hand-drawn flowcharts**

Title chains get redrawn in Visio or Miro for every deliverable.

- **Float-math drift**

Decimals don't roll up cleanly. Pennies get lost. Auditors push back.

- **No audit trail**

When someone asks "where did 0.0234375 come from," the answer is in the landman's head.

“

The math is right. The proof of the math is not.

— Every landman on a deadline

OVERVIEW

One workspace for the title chain, the math, and the paper.

LANDroid is a desk-side application for mineral-title work — oil & gas, fee and state leases. Build a chain of title on a visual canvas, let the engine cascade royalty math in exact-precision decimals, attach the underlying documents, and hand a clean deliverable to a broker, partner, or auditor.

9

decimal places of exact precision

1

engine, every burden cascaded

5

core title operations

0

float-math epsilon hacks

Local-first. Decimal-exact. Document-grounded.

Desk Map — the title chain on a canvas.

A Miro-style canvas where every conveyance is a card.

- **Convey, Precede, Attach, Delete**

Five title operations. Each one updates the cascade in real time.

- **Retain & branch indicators**

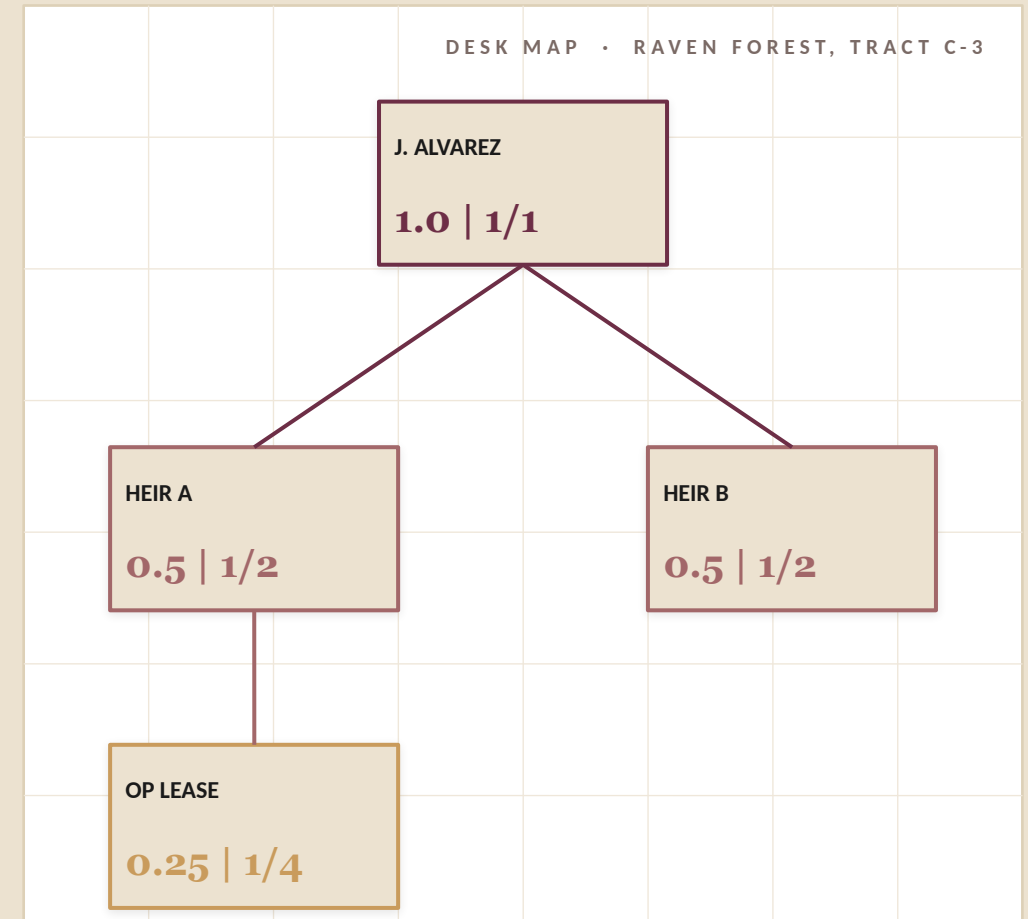
See exactly which fraction stayed with the grantor and which went out.

- **Multi-document attachments**

Drag the deed, the affidavit, and the correction onto a single node.

- **Tract-aware coverage totals**

Found in chain, linked owners, leased — at a glance, per tract.



Exact-precision math that doesn't drift.

Decimal.js

not floats

0.062500000

1/16 — rendered alongside the decimal, every time, by a Stern-Brocot fraction finder.

Five operations. Every cascade guarded.

executeConveyance

grantor → grantee for a fractional interest

executeRebalance

re-proportion siblings after a correction

executePredecessorInsert

splice an heir or missing link into the chain

executeAttachConveyance

re-parent a branch onto a different ancestor

executeDeleteBranch

remove a node and clean cascade downstream

Every decimal shows its work.

Hover any number in the app. The full derivation pops out — inputs, formula, substitutions, result. No more "trust me" math.

Owner Tract Royalty

Formula

Leased Fraction × Royalty Rate

Substituted

$(1/2) \times (1/4)$

Result

0.125000000 · **1/8** · **12.50%**

Why this matters

- Auditors stop asking how you got there
- Brokers can verify decimals without opening the file
- Same shape feeds the printable Audit Sheet export
- Reusable formula objects — engine, UI, and print stay in sync

Leasehold — burdens, splits, unit economics.

The leasehold view computes unit-decimal economics tract by tract: tract participation, leased ownership, every burden stacked in date order, every assignment netted out.

Tract Participation Factor

Per-tract pooled-acre weighting with the formula printed inline. No more spreadsheet TPF cells with stale references.

NPRI Burdens — fixed & floating

Whole-tract, branch, and royalty-basis NPRIs stack correctly. Over-burdened tracts surface as warnings, not silent drift.

ORRI by basis

Gross 8/8, WI, or NRI basis — assigned per-unit and per-tract, layered in chronological order.

WI splits & transfer orders

Pre-WI subtraction shown step by step. Assignments roll up into a transfer-order with variance against expected coverage.

Documents — every paper in one drawer.

Phase 7A. One row per document. Saved views for the work you do all day.

● **Saved views**

Inbox • Runsheet • Leasehold • Curative • Research • GIS • Federal • Duplicates • Needs OCR

● **Metadata you actually use**

Instrument type, county, vol/page, effective & recording dates, grantor/grantee, OCR status

● **Linked-entity display**

See which nodes the document attaches to — and jump straight to the Desk Map

● **Content-hash dedup**

Drop the same PDF twice — LANDroid catches it

● **Packet builder**

Filter to a view, preview the packet, export a JSON manifest

DOCUMENT	AREA	DATE
Alvarez → Heirs WD	Runsheet	1962
NPRI Reservation	Curative	1978
Op Lease (Texaco)	Leasehold	1981
Affidavit of Heirship	Curative	1994
Correction Deed	Runsheet	2011
BLM Lease 56-A	Federal	2019

AI co-pilot — bounded, grounded, sandboxed.

LANDroid uses AI where it earns its keep — turning messy spreadsheets into nodes, and answering domain questions from your workspace. Not for inventing title.

SPREADSHEET IMPORT WIZARD

- Drop an .xlsx or .csv — parsed in a sandboxed Web Worker
- 256KB / 32 sheets / 5000 cells parse caps
- Model proposes sheet classification + column mapping
- Schema-enforced output — no hallucinated fields
- You review every row before it touches the canvas

CHAT ASSISTANT

- Domain-tuned system prompt — no invented citations
- Hosted mode is read-only — answers, doesn't mutate
- Local mode unlocks ~25 supervised authoring tools
- Per-user 500K daily token ceiling enforced server-side
- Keys session-only; hosted requests proxied via Lambda

Built for the working landman.

1 Owners

Mineral fraction, NMA, entity type, county, deceased flag, probate venue, contact.

2 Curative

Track title issues — heirship, missing heirs, stipulations — through to resolution.

3 Research

Project notes, RRC imports, saved questions. Parking lot for everything you read.

4 Federal Leasing

BLM lease tracking with communitization references. Reference-side today.

5 Maps

Project map assets, regions, and external GIS references kept with the workspace.

6 Demo Data

Raven Forest (combinatorial) and Crackbaby Carnival (planted errors) for training.

Deliverables a broker will actually open.

The output side. Everything below shipped or actively in flight.

SHIPPED

Printable flowchart

The Desk Map renders to a print overlay. Browser save-as-PDF, ready for a title-opinion exhibit or a broker handoff.

SHIPPED

.landroid round-trip

Full workspace, owners, leases, documents, and metadata — exported and re-imported with schema-versioned migration (v7 → v8).

IN FLIGHT

Audit Sheet export

One page per tract — every input, every derived number, every formula printed inline. Tooltips, on paper. Designed for the auditor who can't sit at the screen.

STACK

Vite 6 · TypeScript strict · React 18 · React Flow · Decimal.js

Dexie · Zustand · Tailwind v4 · Vitest + Playwright · Vercel AI SDK

Decimal-exact title. Document-grounded. Desk-side ready.

582

unit tests passing

11

Playwright e2e workflows

v8

.landroid schema

